2016 EXAMINATIONS

Lancaster University

**Part II**

**COMPUTING AND COMMUNICATIONS**

**SCC.341 Game Design [2.5 Hours]**

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*Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.*

***Use a separate answer book for each question.***

## Question 1

**1.a** Huizinga has described the concept of play as follows,

*"Summing up the formal characteristic of play, we might call it a free activity standing quite consciously outside 'ordinary' life as being 'not serious' but at the same time absorbing the player intensely and utterly. It is an activity connected with no material interest, and no profit can be gained by it. It proceeds within its own proper boundaries of time and space according to fixed rules and in an orderly manner. It promotes the formation of social groupings that tend to surround themselves with secrecy and to stress the difference from the common world by disguise or other means."*

Identify the similarities between the above statement and Robert Callois' categorization of play into Agon, Alea, Mimicry and Ilinx across Paidia and Ludus. Use examples of games to explain your arguments.

**[12 marks]**

**1.b** The relationship between a game's decision and a game's outcome can have two degrees of uncertainty: a risk outcome and an uncertain outcome.

Using a game of your choice that contains both degrees of uncertainty:

i. Briefly describe the game and list its operational rules.

**[3 marks]**

ii. Explain how these degrees of uncertainty can be manifested in your game.

**[4 marks]**

iii. Identify risk or uncertainty have an inherent connection with meaningful play.

**[1 marks]**

iv. Describe how this inherent connection with meaningful play arises.

**[5 marks]**

**Total 25 marks**

## Question 2

**2a.** Contrast the categorization of women gamers as identified by marketing research versus stereotypical (or anecdotal) categorization.

**[8 marks]**

**2b.** As a game developer or a game admin, how would you control/curate the occurrence of negative social interactions like misogynistic behaviour and bullying?

**[5 marks]**

**2c.** Consider the example of 'Cake Division' and its optimal solution where each player maximizes their self-interest by letting the first player divide the cake and the second player choose which of the two pieces he wants.

i. Map the decision making process in a matrix and label all relevant axes.

**[4 marks]**

ii. Deduce the utility values from the decision matrix above and use these to label the pay-off matrix. **[2 marks]**

**2d.** Discuss how a game designer can use games as game theory systems to maximize their self-interest (more users) against potential conflicts arising out of misogynistic actions of gamers.

**[6 marks]**

**Total 25 marks**

## Question 3

**3a.** Provide the definition of meaningful play in descriptive & evaluative contexts. **[6 marks]**

**3b.** Using the evaluative definition of meaningful play identify the game elements within the scene below that point to meaningful play. **[4 marks]**

**3c.** Systems fall into one of the four following categories: Fixed, Periodic, Complex, or Chaotic.

i. Briefly describe the characteristics associated with each category. **[2 marks]**

ii. Which of these systems contribute to and enable meaningful play? Using a game as an example, give your rationale on how meaningful play can attained.

**[4 marks]**

**3d**. Consider a game of your choice that falls into the category of games as systems of information.

Describe how this game will use and manipulate information.
In your description, consider how the information is known to all players, hidden, randomized, acquired and transformed throughout the game.

**[9 marks]**

**Total 25 marks**

## Question 4

**4.a** Design/sketch an outline of a single-player smartphone based game. Describe how your proposed game satisfies the design principles listed below. When designing your game consider also its rules based on the following three levels: operational rules, constitutive rules and implicit rules.

***Design Principles by Shigeru Miyamoto:***

- *Start with a simple concept*
- *Design around the system's limitations*
- *Minimize the player's confusion*
- *Test gameplay*
- *Incorporate a smooth learning curve*
- *Accommodate all skill levels* **[15 marks]**

**4.b** Based on the game you designed in 4.a above, describe its rules for each of the three levels listed below:

i. Operational rules. **[3 marks]**

ii. Constitutive rules. **[2 marks]**

iii. Implicit rules. **[2 marks]**

**4.c** How does your game exhibit its identity and uniqueness? **[3 marks]**

**Total 25 marks**

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